DM – Generalized permutations and combinations

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Outline

1. Repeating objects
2. Identical objects
3. Balls and boxes
   - Distinguishable balls, distinguishable boxes
   - Identical balls, distinguishable boxes
   - Distinguishable balls, identical boxes
   - Identical balls, identical boxes

Permutations with repetition



How many different outcomes are there when three dice are cast, if the order of the dice matters?

Ans: $6 \times 6 \times 6 = 216$ outcomes (by product rule).

Theorem

The number of r-permutations of n elements with repetition allowed is n^r.

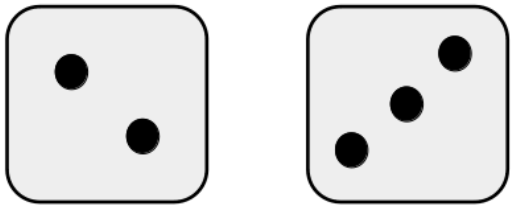
Proof: By the product rule.

Example

Given two positive integers m and n, how many functions are there from $\{1, 2, \ldots, m\}$ to $\{1, 2, \ldots, n\}$?

Ans: Such a function is completely defined by an m-tuple $\{f(1), f(2), \ldots, f(m)\}$ where $1 \leq f(i) \leq n$.
There are n^m such m-tuples.

Combinations with repetition



*How many different outcomes are there when two dice are cast, if the order of the dice do not matter (*i.e. $(1, 2)$ *and* $(2, 1)$ *are the same outcome)?*

Method 1: List all possible outcomes and count.

Method 2:

- There are $C(6, 2) = 15$ outcomes where the two dice yield different numbers.
- There are $C(6, 1) = 6$ outcomes where the two dice yield the same number.

Ans: 21.

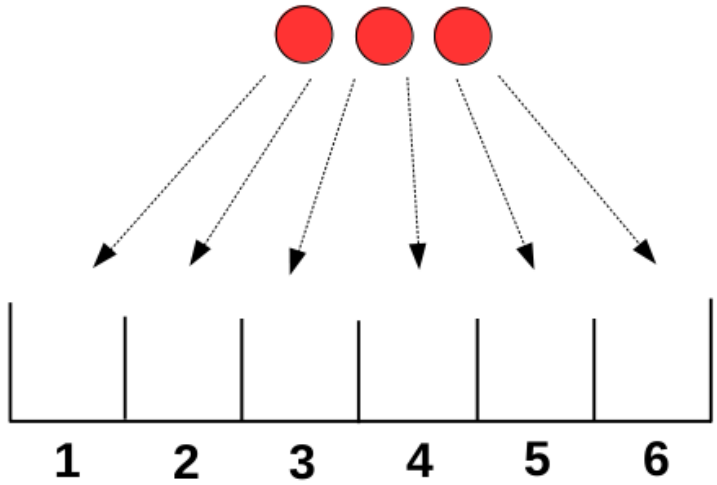
Combinations with repetition



How many different outcomes are there when three dice are cast, if the order of the dice do not matter?

Both previous methods seem not useful in this case!

Idea: Instead of casting three dice, consider casting three *identical* balls into six boxes numbered from 1 to 6.



Combinations with repetition

Idea: Instead of casting three dice, consider casting three *identical* balls into six compartments numbered from 1 to 6.

Examples:



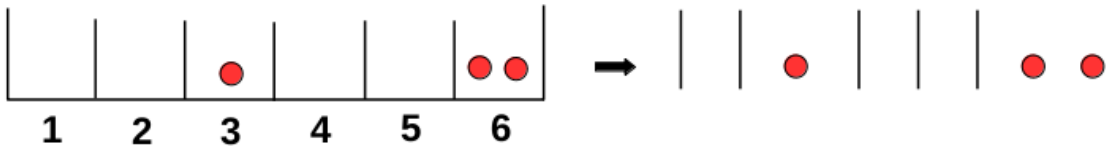


How many different ways are there to put three identical *balls into six* different *compartments?*

Combinations with repetition

How many different ways are there to put three identical *balls into six* different *compartments?*

A further observation: Each way to put the balls into the compartments is *uniquely* determined by an arrangement of three balls and five separators in a row!



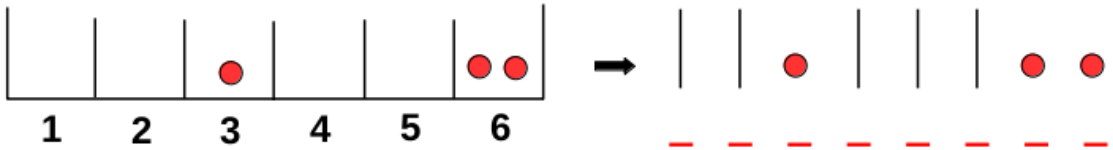
The number of such arrangements is the number of ways to choose three out of $3 + 5 = 8$ places for the balls!

Ans: $C(8, 3) = 56$.

Combinations with repetition

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Ans: $C(8, 3) = 56$.

Combinations with repetition

Theorem

The number of r-combinations of n elements when repetition is allowed is $C(n+r-1, r) = C(n+r-1, n-1)$.

Proof idea: Consider each object a "ball" and each element type a "compartments". (*Note:* The balls are identical; the compartments are different.)
Each way to put the balls into the compartments corresponds to one and only one way to arrange the r balls and $n-1$ separators in a row, which is $C(n+r-1, r)$.

Examples

A cookie shop has four different kinds of cookies. How many different ways can six cookies be chosen? (Assume that only the type of cookie, and not the individual cookies or the order in which they are chosen, matters.)

Each way is a 6-combination of four types of cookies with repetition allowed.

Ans: $C(6+4-1,6) = C(9,6) = 84$ ways.

Examples

How many non-negative integer solutions are there for the equation

$$x_1 + x_2 + x_3 = 11\,?$$

Each solution corresponds to a way to distribute 11 units into three variables x_1, x_2, x_3. We have $r = 11$, $n = 3$.
Ans: $C(11 + 3 - 1, 11) = C(13, 11) = 78$ ways.

Permutations with identical objects

How many strings can be made by shuffling the letters of SUCCESS*?*

Method 1:

- There are 7! permutations of the seven letters.
- Each permutation is repeated $3! \times 2!$ times by permuting the three S's and/or the two C's.
- By the division rule, there are $7!/(3!2!1!1!) = 420$ different strings.

Permutations with identical objects

How many strings can be made by shuffling the letters of SUCCESS*?*

Method 2:

- **Step 1:** Choose three out of seven places for the S's; there are $C(7,3)$ ways.
- **Step 2:** Choose two out of four remaining places for the C's; there are $C(4,2)$ ways.
- **Step 3:** Choose one of the two remaining places for the U; there are $C(2,1)$ ways.
- **Step 4:** There is $C(1,1)$ choice left for the E.
- In total, there are $C(7,3) \times C(4,2) \times C(2,1) \times C(1,1) = 35 \times 6 \times 2 \times 1 = 420$ different strings.

Combinations with repetition

Theorem

The number of different permutations of n objects, among which are n_1 identical objects of type 1, n_2 identical objects of type 2, etc. is

$$\frac{n!}{n_1!n_2!\ldots n_k!}.$$

Proof: Consider the following procedure to form such a permutation:

- **Step 1:** Choose n_1 positions among the n available positions for the type 1 objects, there are $C(n, n_1)$ choices;
- **Step 2:** Choose n_2 positions among the $n - n_1$ remaining positions for the type 2 objects, there are $C(n - n_1, n_2)$ choice;

  ...

In total, there are $C(n, n_1) \times C(n - n_1, n_2) \times \cdots \times C(n_k, n_k) = \frac{n!}{n_1!n_2!\ldots n_k!}$ choices.

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Distinguishable balls, distinguishable boxes

Theorem

The number of ways to distribute n distinguishable balls into k distinguishable boxes so that n_i objects are placed into box i, $i = 1, 2, ..., k$ is

$$\frac{n!}{n_1! n_2! \dots n_k!}.$$

(**Note:** In this scenario, the number of balls in each box is fixed.)

Example

At a football tournament, the 32 participating teams are to be divided into 8 groups of four. Ignoring restrictions such as seeding, how many possible group draw results are there?

Ans:

$$\frac{32!}{(4!)^8} \approx 2.39 \times 10^{24}.$$

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Identical balls, distinguishable boxes

Theorem

The number of ways to put r identical balls into n distinguishable boxes is

$$C(r+n-1, r) = \frac{(r+n-1)!}{r!(n-1)!}\,.$$

(**Note:** In this scenario, there is no restriction on the number of balls in each box.)

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Distinguishable balls, identical boxes

How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

List and count:

- $(4, 0, 0)$: 1 way.
- $(3, 1, 0)$: 4 ways.
- $(2, 2, 0)$: 3 ways (why is it not 6?)
- $(2, 1, 1)$: 6 ways (why is it not 12?)

Ans: 14.

Distinguishable balls, identical boxes

Definition

The number of ways to put n distinguishable balls into k identical boxes so that every box is not empty is called the Stirling numbers of the second kind and is denoted by $S(n, k)$.

Theorem

$$S(n,k) = \frac{1}{k!}\sum_{i=0}^{k-1}(-1)^i\binom{k}{i}(k-i)^n\,.$$

Corollary

The number of ways to put n distinguishable balls into k boxes (some of which may be empty) is

$$\sum_{j=1}^{k} S(n,j)\,.$$

Distinguishable balls, identical boxes

How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

- 1 non-empty office: $S(4,1) = 1$.
- 2 non-empty office: $S(4,2) = 7$.
- 3 non-empty office: $S(4,3) = 6$.

Total: $1 + 7 + 6 = 14$ ways.

Stirling numbers of the second kind

Theorem

For all positive integers n, k such that $n \geq k$:

$$S(n+1,k) = kS(n,k) + S(n,k-1)\,.$$

Proof: By double-counting the number of ways to partition $n+1$ distinguishable balls into k non-empty identical boxes.

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Identical balls, identical boxes

A way to put n identical balls into k identical boxes corresponds to a partition *of n into k positive integers, that is one way to write $n = a_1 + a_2 + \cdots + a_k$, where a_i are positive integers and $a_1 \geq a_2 \geq \cdots \geq a_k$.*

- The number of partitions of a positive integer n is represented by the *partition function* $p(n)$.
- There is no known closed-form formula for $p(n)$.
- Approximation formulas exist (Hardy and Ramanujan; Rademacher).